

Practical work with a spectrometer

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Outline

PURPOSES:

Help understanding the theory

The aims of recorded data

Mechanism of spectrometer for remote sensing

I. Background and definitions

1.1 Radiometric variables

1.2 From radiance to reflectance / surface

1.3 Application to satellite ground data

II. Spectral measurement

III. Practical works

2.1 Introduction equipment

2.2 Test equipment

2.3 Steps to do the experiment

2.4 Measuring spectral reflectance by yourself

I. Background and definitions

Blackbody radiation, **total** emitted radiation [W/m²] :

$$M = \sigma \cdot T^4$$

Stefan-Boltzmann constant $5.67 \times 10^{-8} \text{ W/m}^2/\text{K}^4$

Wien's displacement law:

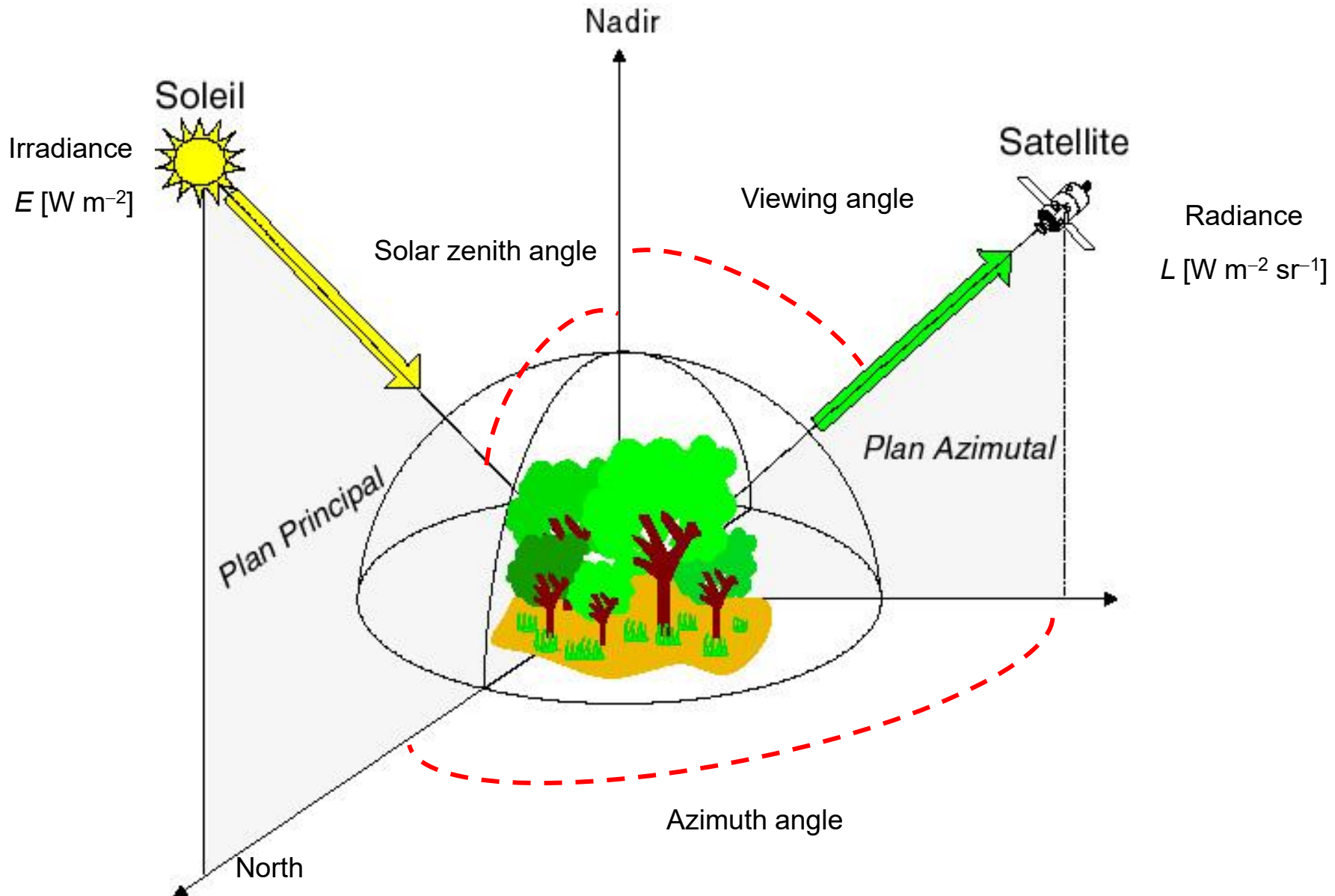
$$\lambda_{\text{max}} \cdot T = b$$

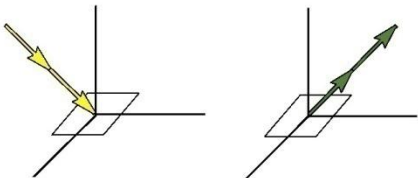
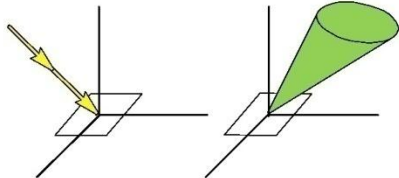
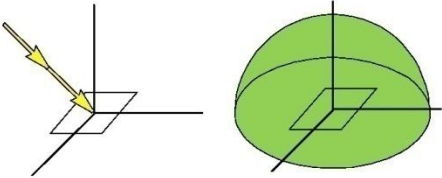
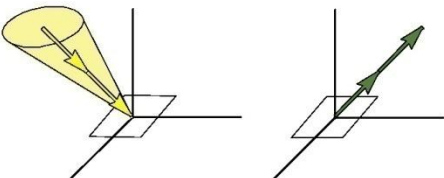
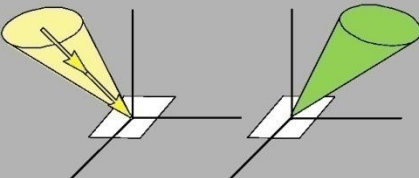
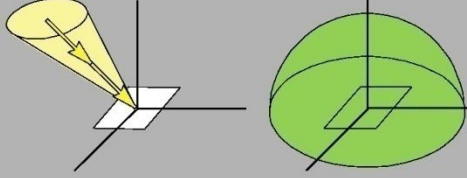
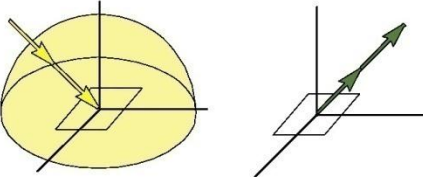
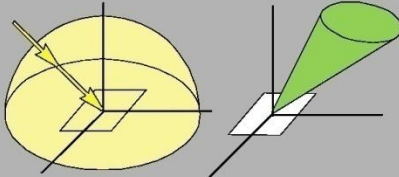
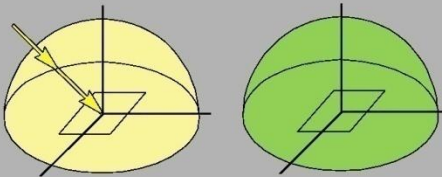
Constant $b = 2,898 \text{ } \mu\text{m} \cdot \text{K}$

Radiometric quantities and units

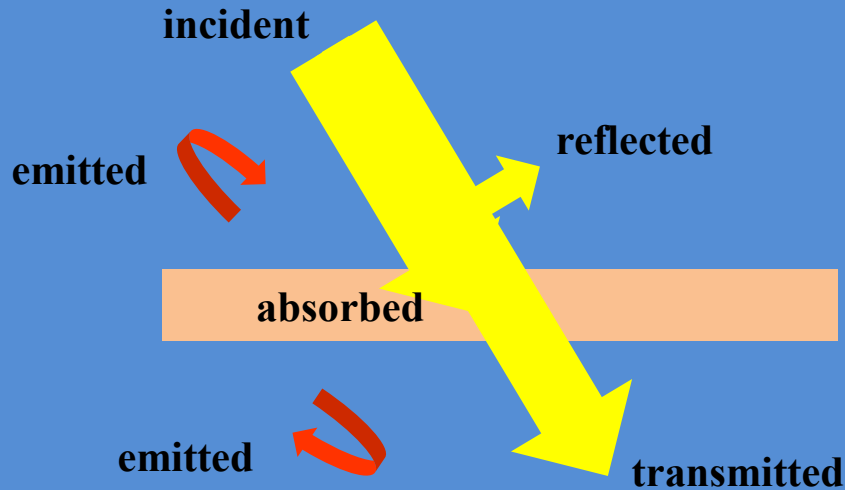
Quantity	Symbol	Unit	Definition
Radiant energy	Q	J	Energy transported by electromagnetic radiation
Radiant flux	Φ	$W=J/s$	Radiant energy per unit time
Irradiance	E	W/m^2	Radiant flux incident on a surface by unit area
Spectral irradiance	E_λ	$W/m^2/nm$	Irradiance at a single wavelength
Radiance	L	$W/m^2/sr$	Radiant intensity emitted by a unit area of an emitting surface
Spectral radiance	L_λ	$W/m^2/sr/nm$	Radiance at a single wavelength
Radiant exitance	M	W/m^2	Radiant flux per unit area that emerges from a surface
Radiant intensity	I	W/sr	Radiant flux radiated into a unit solid angle in a particular direction

Remote sensing system



<i>Incoming/Reflected</i>	Directional	Conical	Hemispherical
<i>Directional</i>	Bidirectional Case 1 	Directional-conical Case 2 	Directional-hemispherical Case 3 
<i>Conical</i>	Conical-directional Case 4 	Biconical Case 5 	Conical-hemispherical Case 6 
<i>Hemispherical</i>	Hemispherical-directional Case 7 	Hemispherical-conical Case 8 	Bihemispherical Case 9 

In the laboratory: reflectance/transmittance/absorptance



Reflection coefficient
or reflectance (DHRF)

$$R(\lambda) = \frac{E_{reflected}(\lambda)}{E_{incident}(\lambda)}$$

Absorption coefficient
or absorptance

$$A(\lambda) = \frac{E_{absorbed}(\lambda)}{E_{incident}(\lambda)}$$

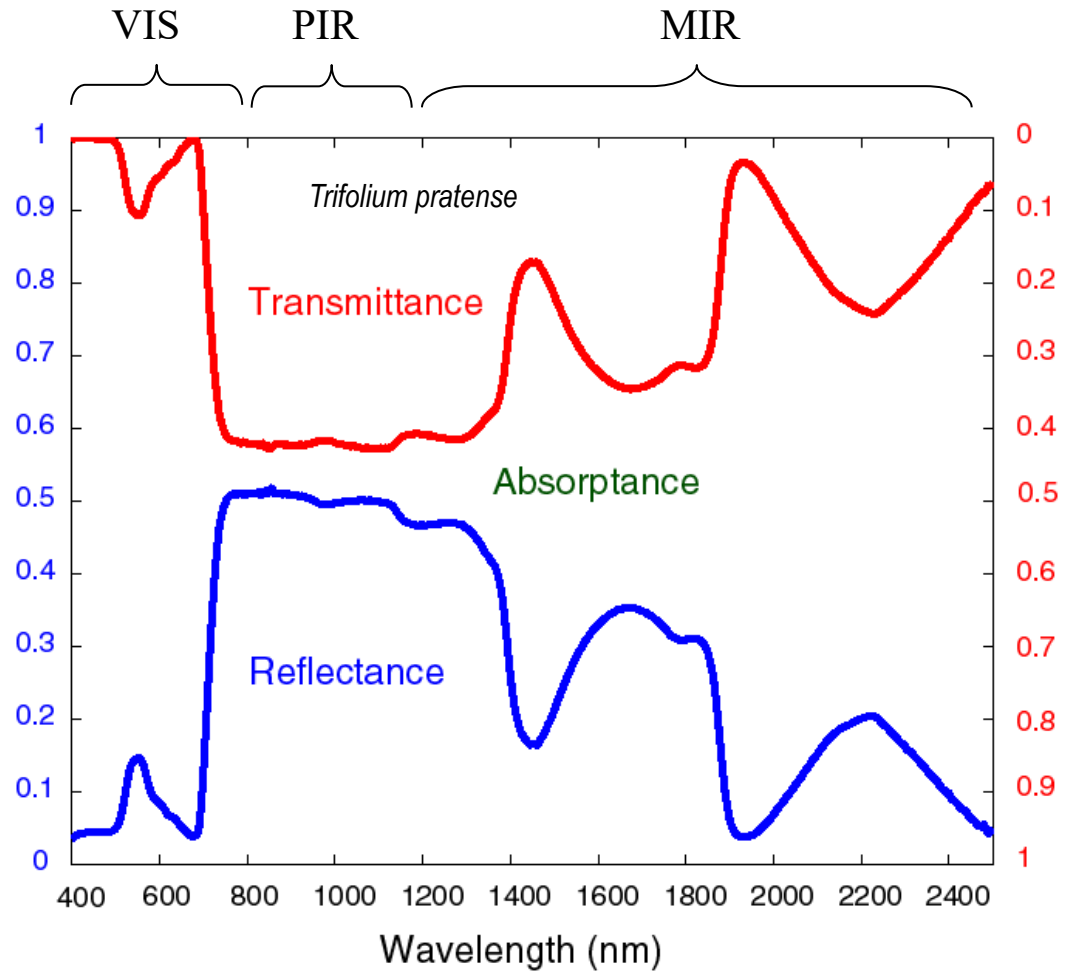
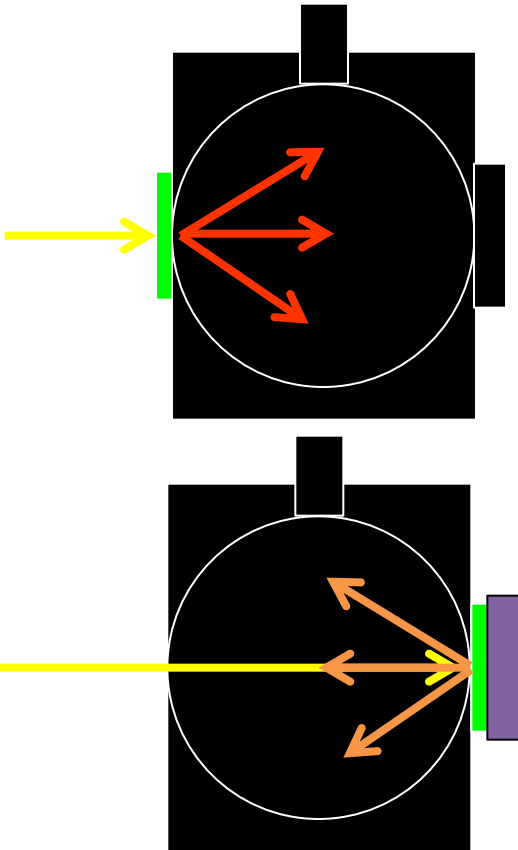
Transmission coefficient
or transmittance (DHTF)

$$T(\lambda) = \frac{E_{transmitted}(\lambda)}{E_{incident}(\lambda)}$$

Law of conservation of energy:

$$R(\lambda) + T(\lambda) + A(\lambda) = 1$$

Vegetation: leaf optical properties



II. Spectral measurement

Instrument

Setting up system

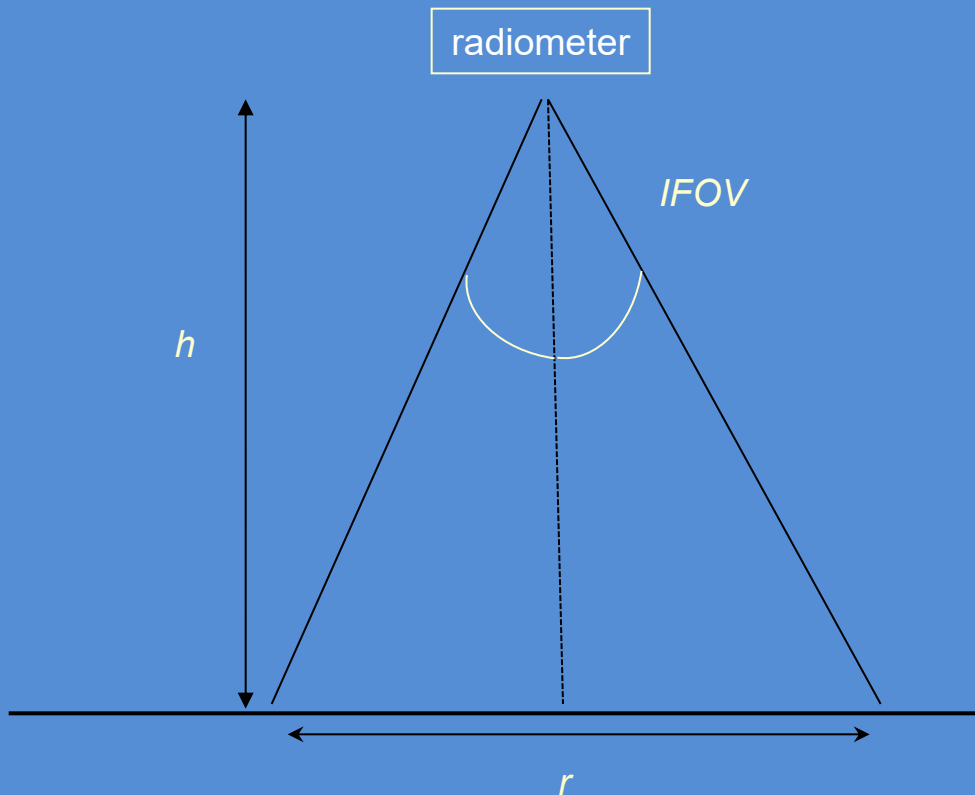
Applications

Instantaneous field of view (IFOV)

IFOV = solid angle through which a detector is sensitive to radiation.

In a scanning System this refers to the solid angle subtended by the detector when the scanning motion is stopped.

The instantaneous field of view is commonly expressed in milliradians



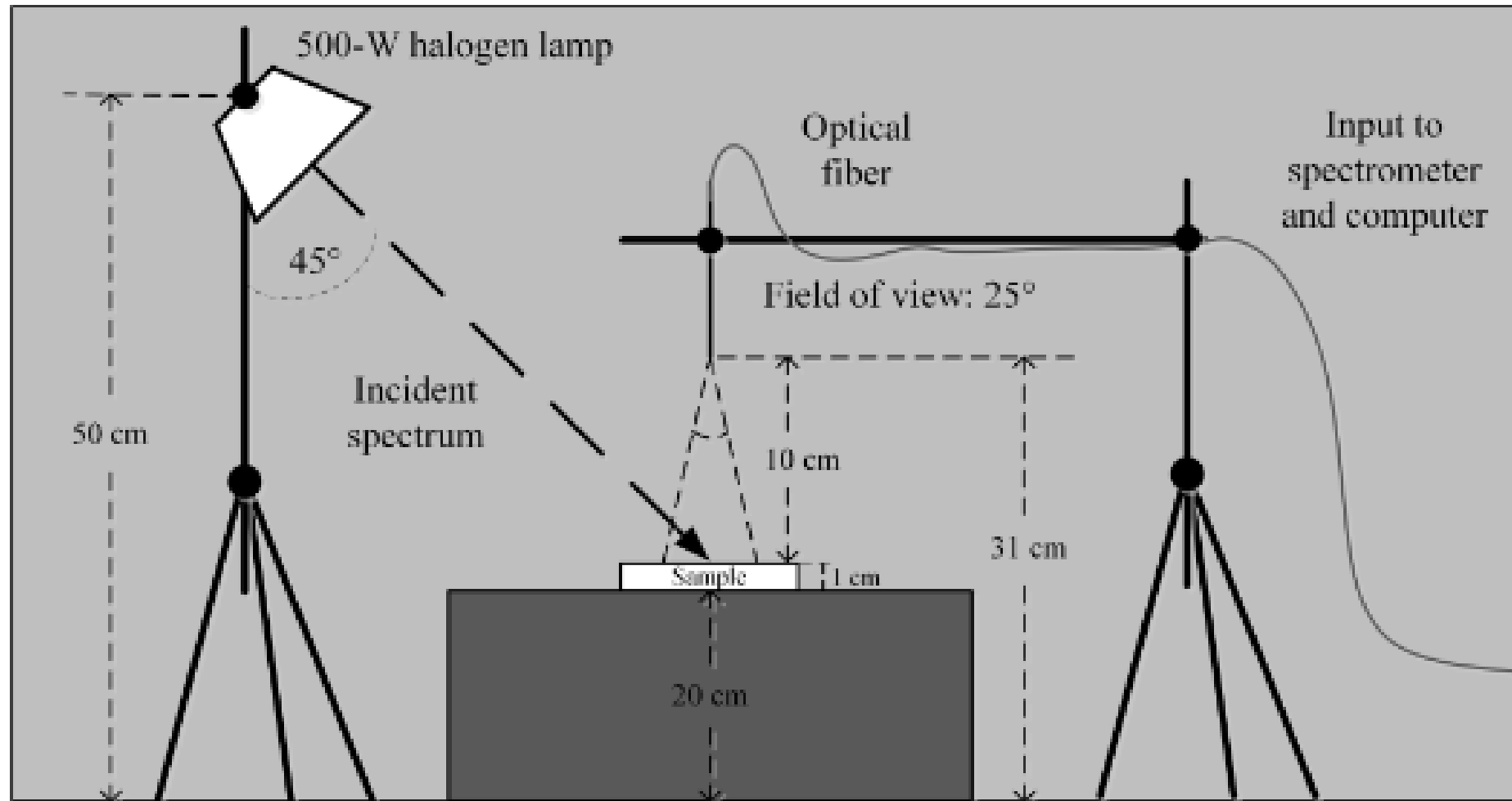
$$\tan \frac{IFOV}{2} = \frac{r}{2h}$$

If small *IFOV* value

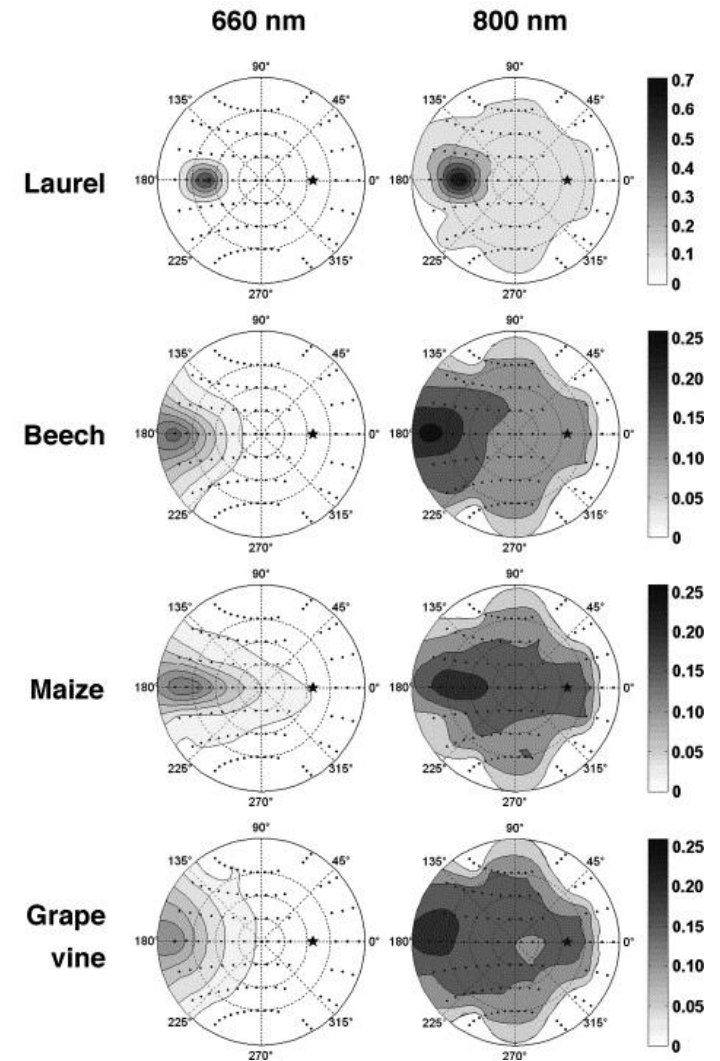
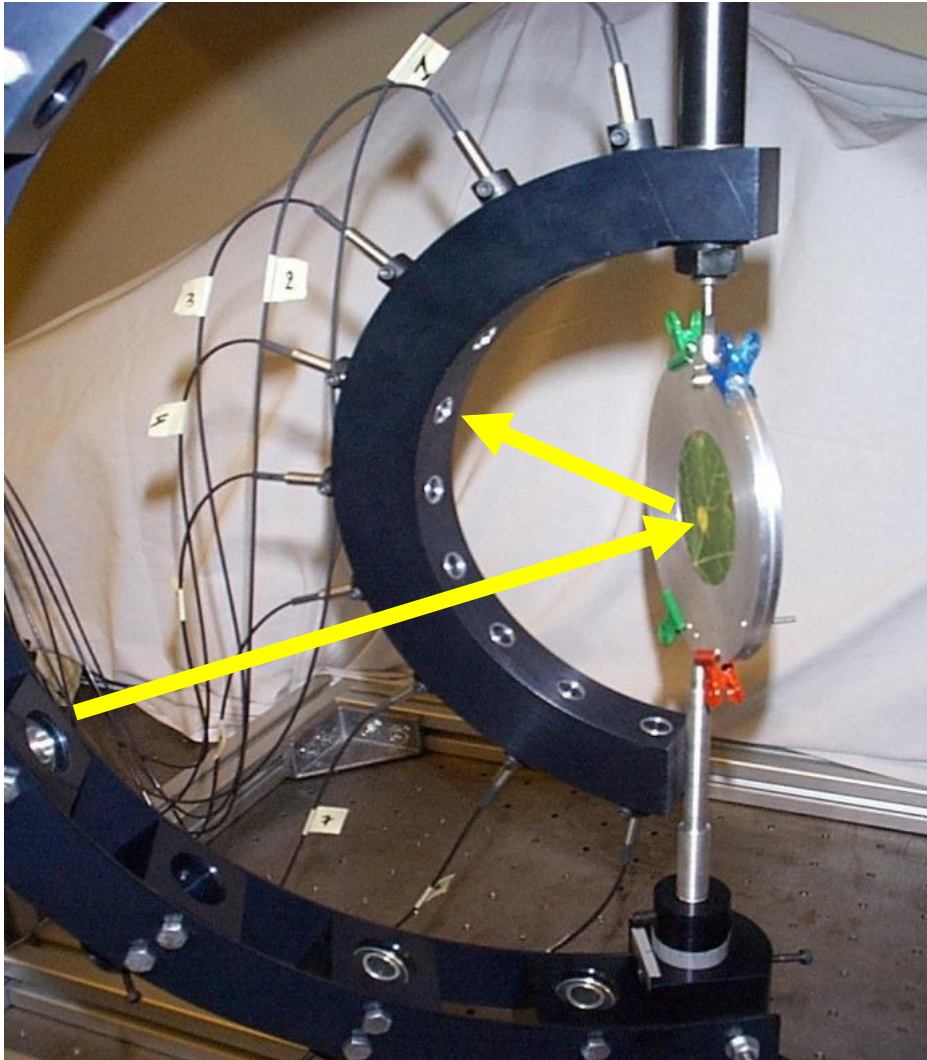
$$\tan \frac{IFOV}{2} \approx \frac{IFOV}{2} = \frac{r}{2h}$$

$$\Rightarrow r = h \times IFOV$$

Setting up the system



In the laboratory: directional measurements using a spectrogoniophotometer



In the field: hand held spectroradiometers

a



b



c



d

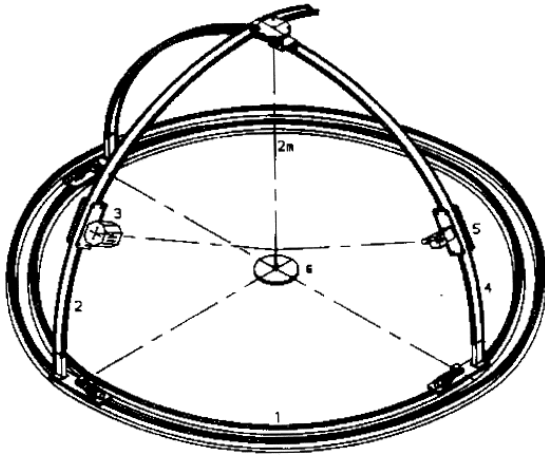


e



In the field: directional measurements using a spectrogoniophotometer

EGO



CLabSpeG

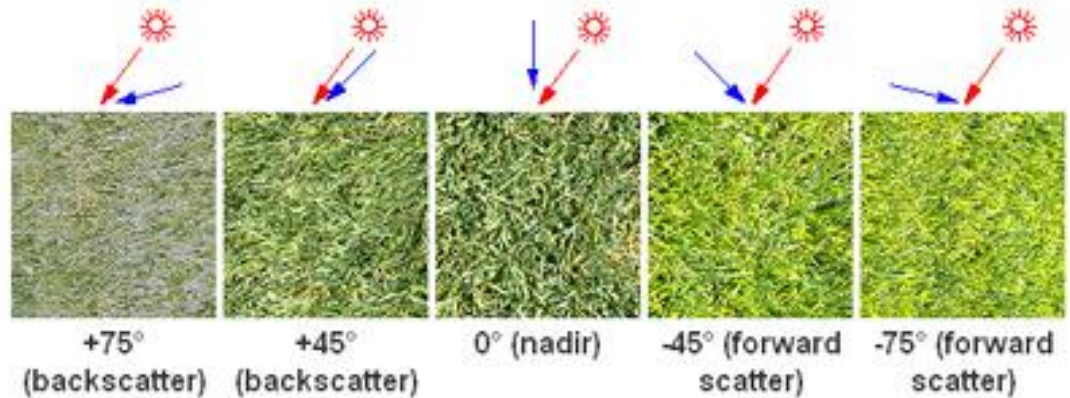


Chamelon

In the field: directional measurements using a spectrogoniophotometer

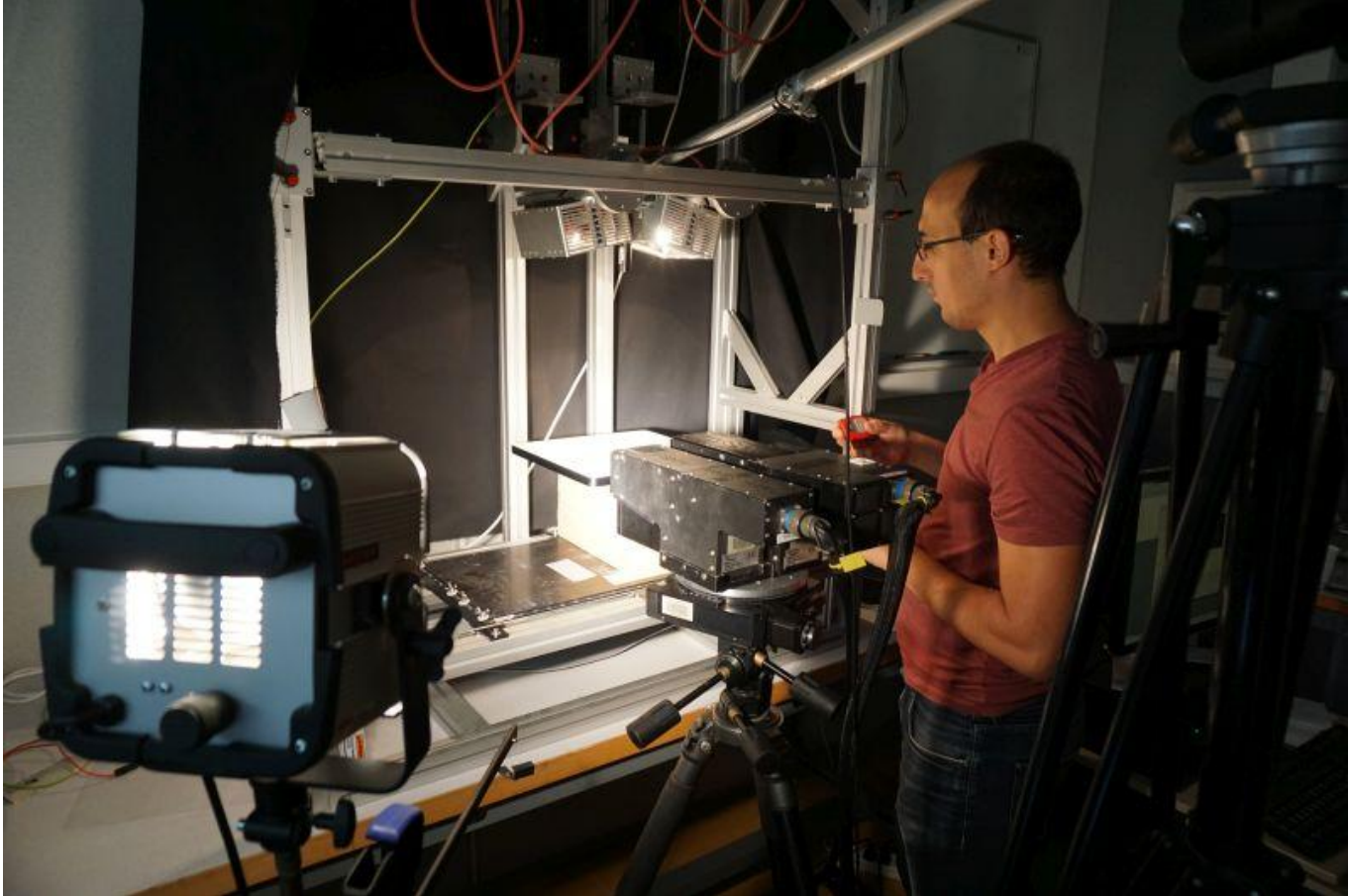


FIGOS goniometer

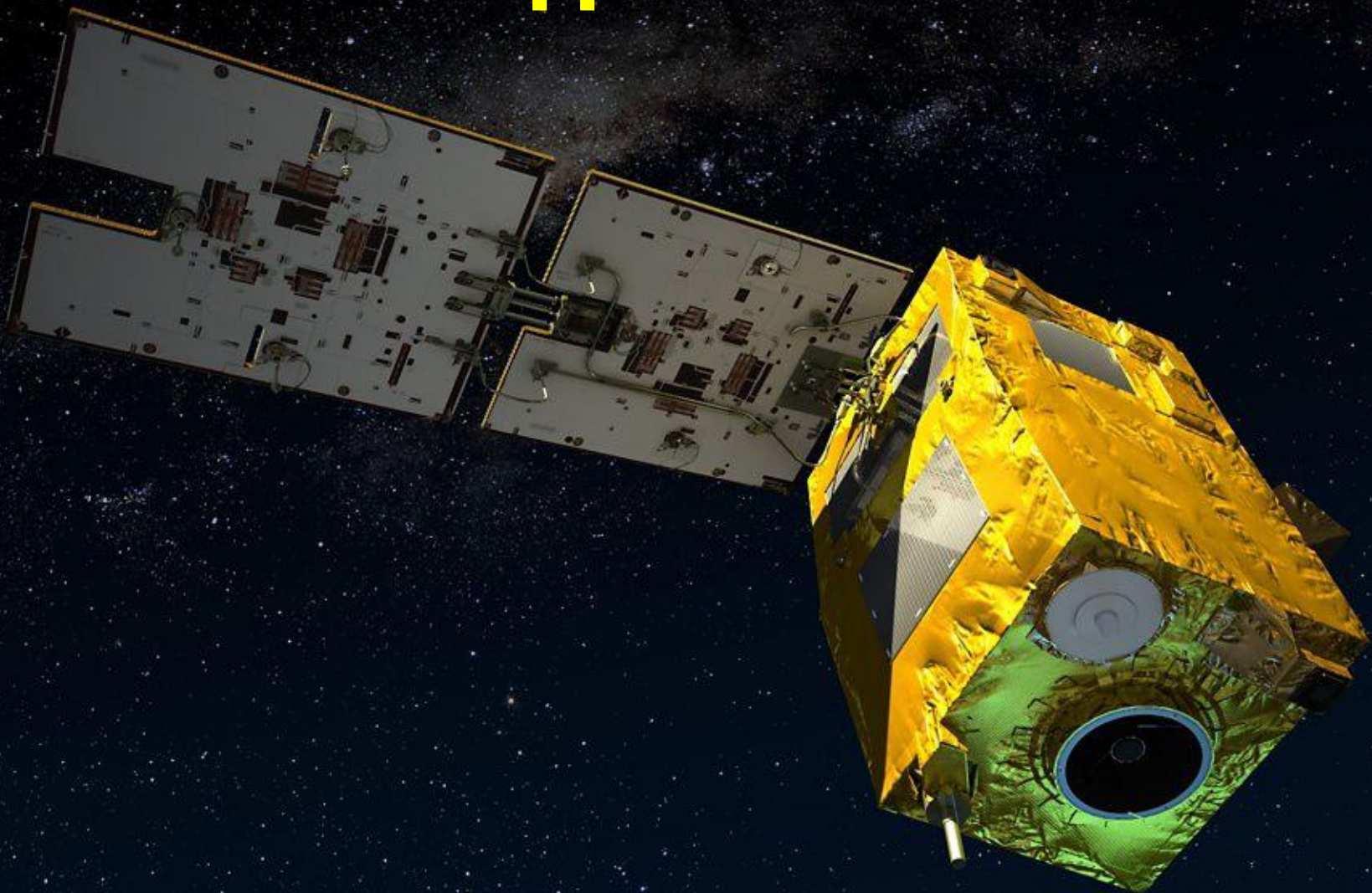


In the laboratory: images using hyperspectral cameras

Caméras Hypspec VNIR-1600 & SWIR-320m-e + spectroradiomètre FieldSpec Pro

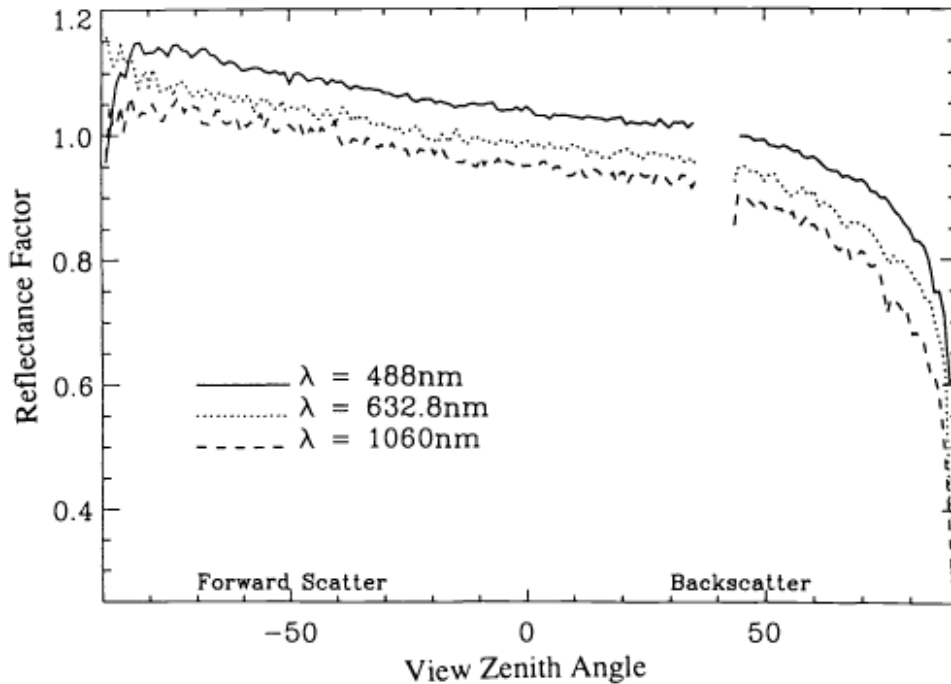


Applications



Satellite calibration

At the satellite level: laboratory calibration of optical sensors (before launch)



Spectralon reflectance factor. Laser illumination at 40 deg provided data at 0.488, 0.6328, and 1.06 μm .



Sentinel-2 LDCM cross calibration campaign

At the satellite level: in flight calibration of optical sensors (after launch)

Irradiance cannot be measured... During the satellite pass, some field optical measurements are performed over calibration sites (cs)

Such calibration sites should have particular properties:

- high and flat spectral reflectance
- invariant reflectance
- spatial uniformity
- accessible site

Note that the radiance L can be converted in any other unit than $W/m^2/sr$: volt, bit-depth, etc.

At the satellite level: in flight calibration of optical sensors (after launch)



White Sands (New Mexico, USA)

At the satellite level: in flight calibration of optical sensors (after launch)



Distribution of World-Wide Radiometric Sites

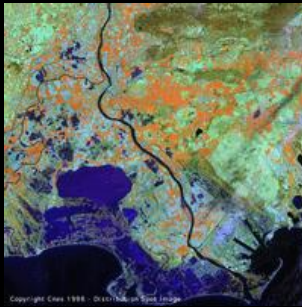
At the satellite level: in flight calibration of optical sensors (after launch)



Plaine de la Crau (France)



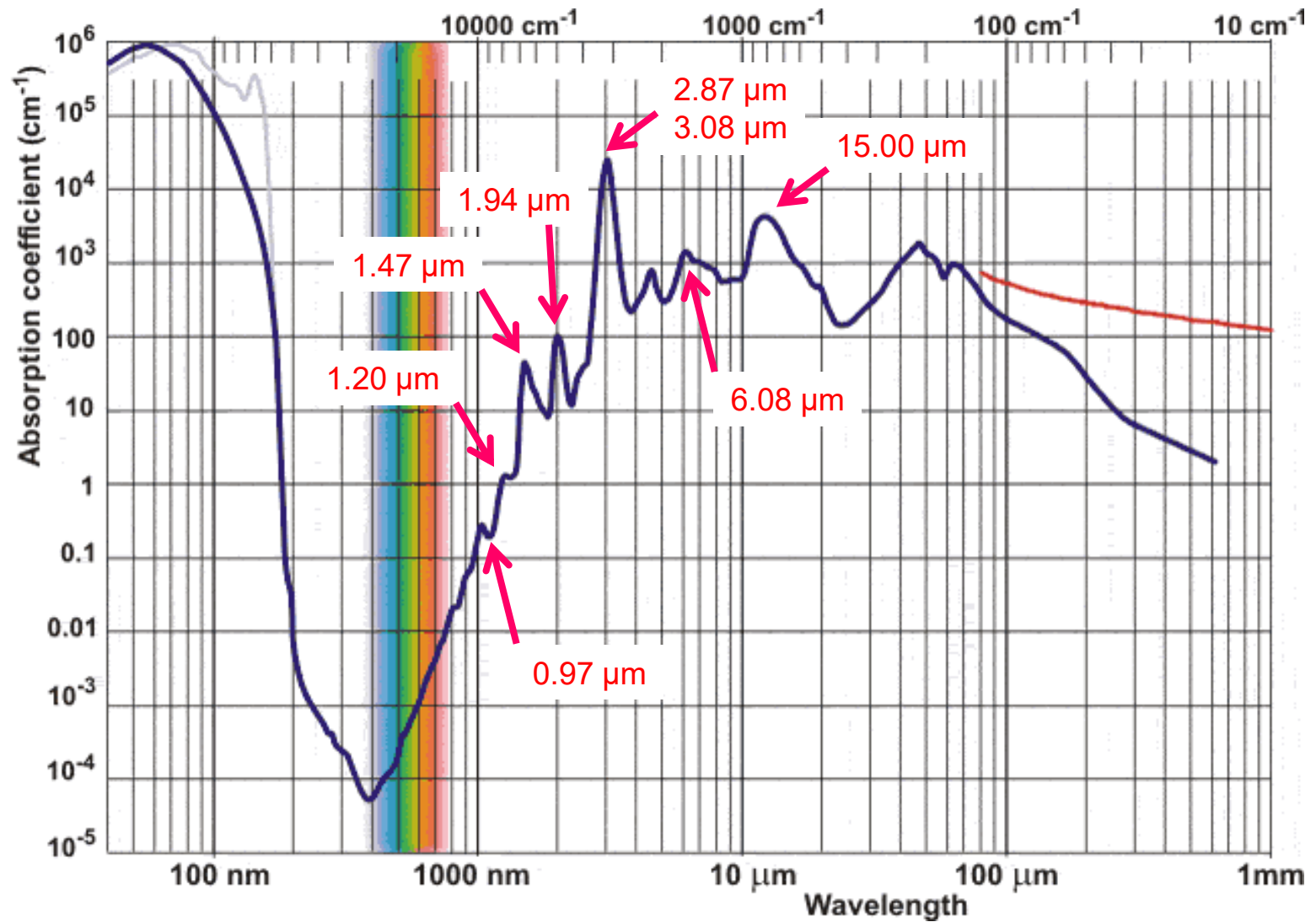
White Sands (Nouveau Mexique, USA)



Surface detection



A common element: the water molecule

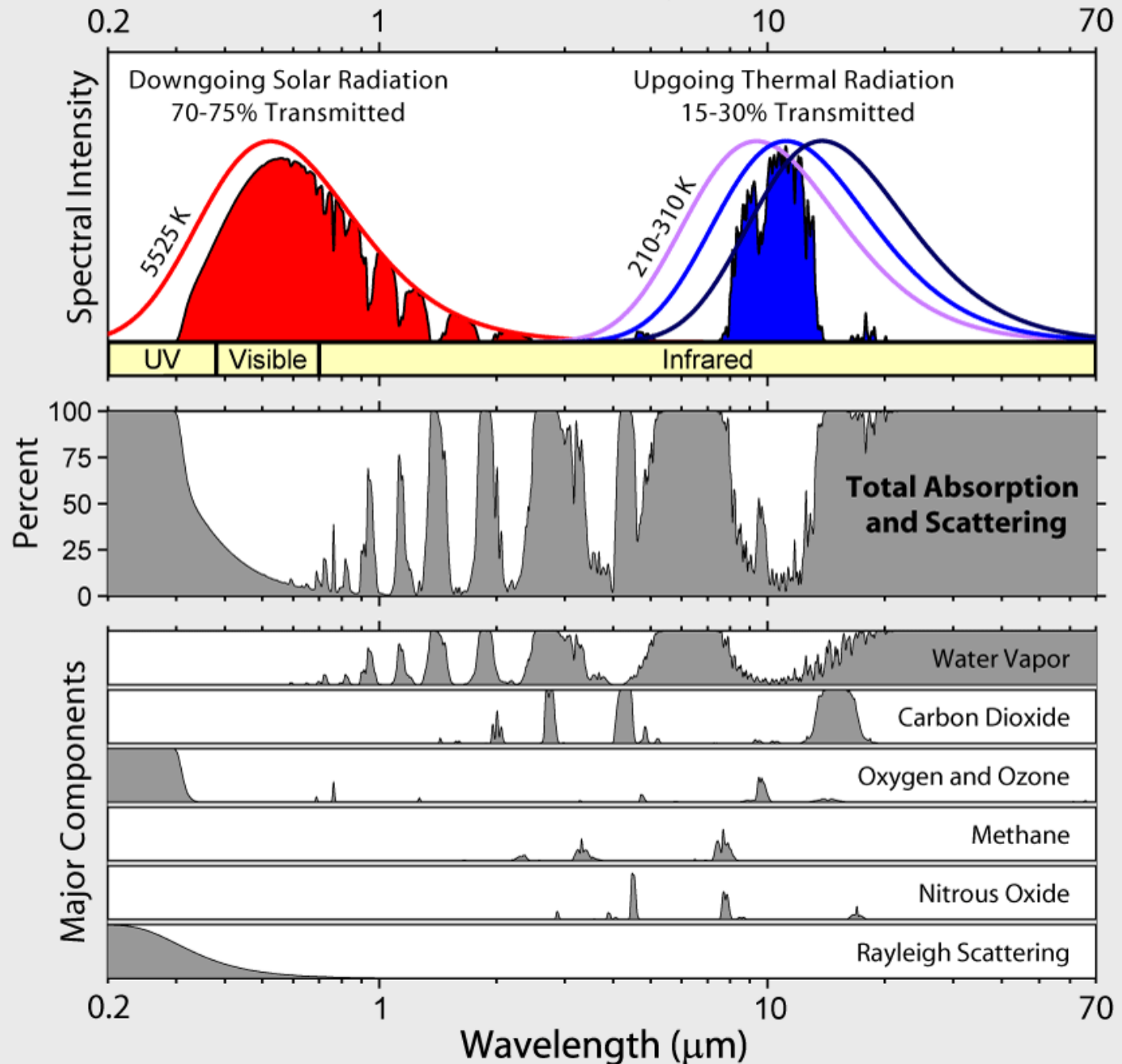


Atmosphere

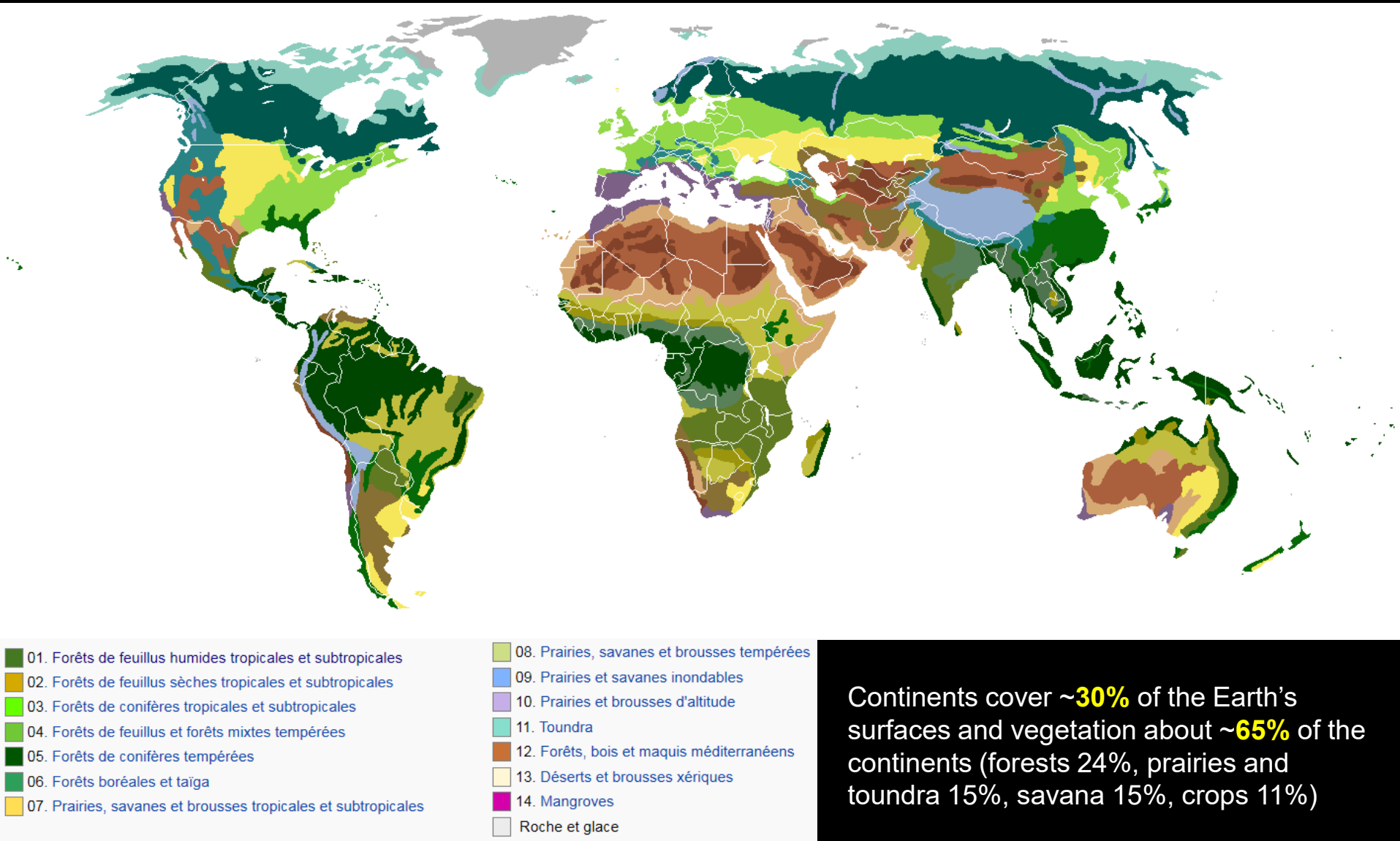
$$\tau = \tau_{\text{abs}} + \tau_{\text{Rayleigh}} + \tau_{\text{Mie}}$$

$$T = T_{\text{abs}} \times T_{\text{Rayleigh}} \times T_{\text{Mie}}$$

Radiation Transmitted by the Atmosphere



Vegetation: main terrestrial biomes

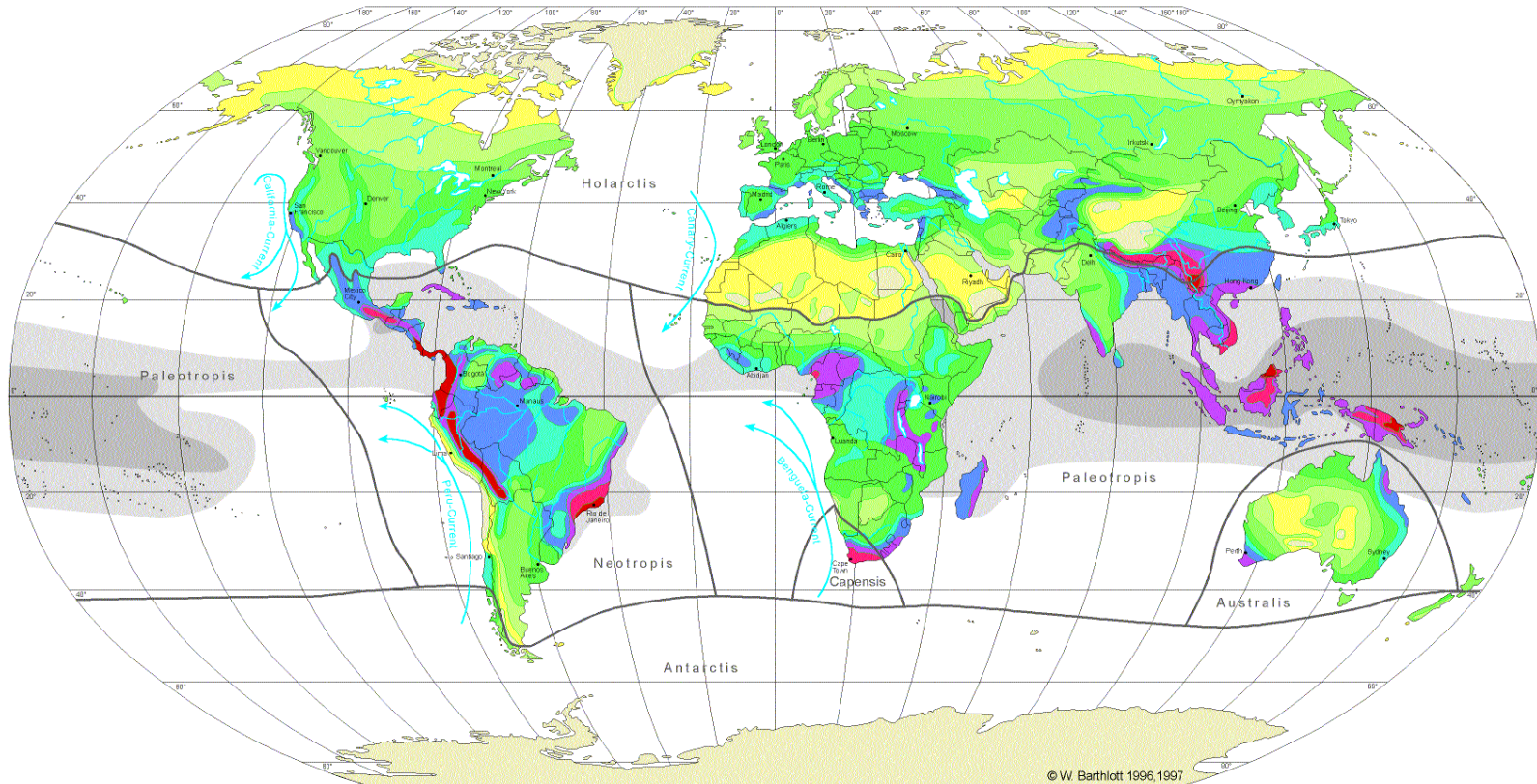


Vegetation: biodiversity classification



Biodiversity is life
Biodiversity is our life

GLOBAL BIODIVERSITY: SPECIES NUMBERS OF VASCULAR PLANTS



Robinson Projection
Standard Parallels 38°N und 38°S
Scale 1: 130 000 000

Diversity Zones (DZ): Number of species per 10.000km²

DZ 1 (<100)	DZ 5 (1000 - 1500)	DZ 9 (4000 - 5000)
DZ 2 (100 - 200)	DZ 6 (1500 - 2000)	DZ 10 (≥5000)
DZ 3 (200 - 500)	DZ 7 (2000 - 3000)	
DZ 4 (500 - 1000)	DZ 8 (3000 - 4000)	

Capensis floristic regions

sea surface temperature

>29°C
>27°C

cold currents

W. Barthlott, N. Biedinger, G. Braun
F. Feig, G. Kier, W. Lauer & J. Mutke 1997
modified after
W. Barthlott, W. Lauer & A. Pläcke 1996
Department of Botany and Geography
University of Bonn
German Aerospace Research Establishment, Cologne

Cartography: M. Gref
Department of Geography
University of Bonn

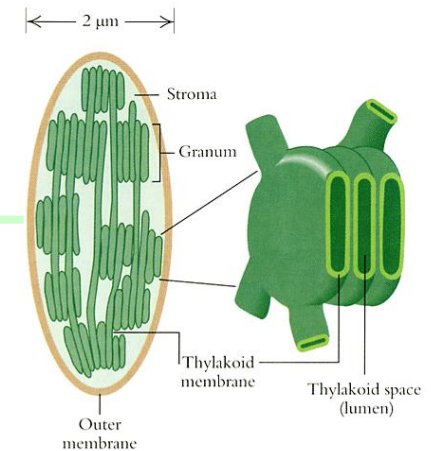
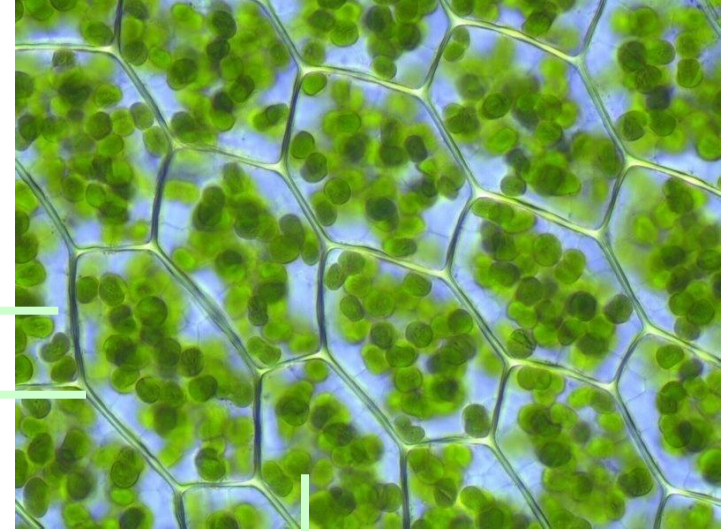
Vegetation: leaf chemical composition

- water (vacuole): 90-95%
- dry matter (cell walls): 5-10%

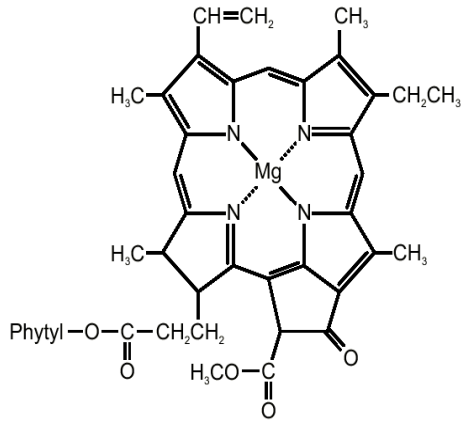
- cellulose : 15-30%
- hemicellulose : 10-30%
- proteins : 10-20% $\Leftrightarrow$ N
- lignin : 5-15% $\Leftrightarrow$ C
- starch: 0.2-2.7%
- sugar
- etc.

- wax(cuticle)
- chlorophylls *a* and *b* + carotenoids (chloroplasts)
- other pigments (cytoplasm)

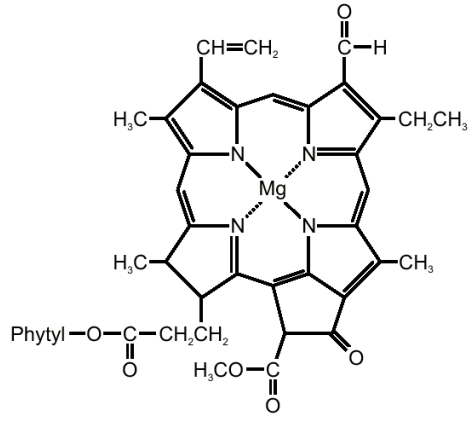
- anthocyanins
- flavones
- brown pigments
- etc.



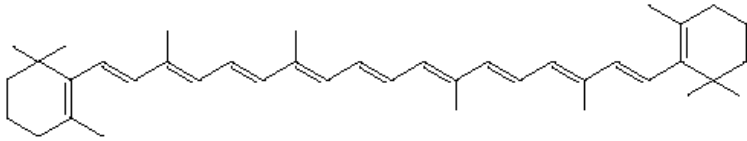
Vegetation: foliar pigments



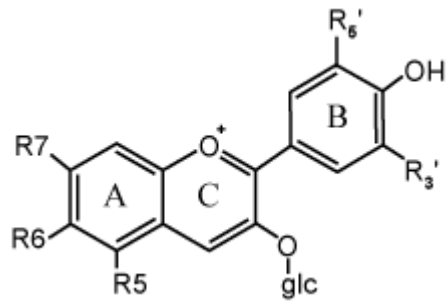
chlorophyll *a*



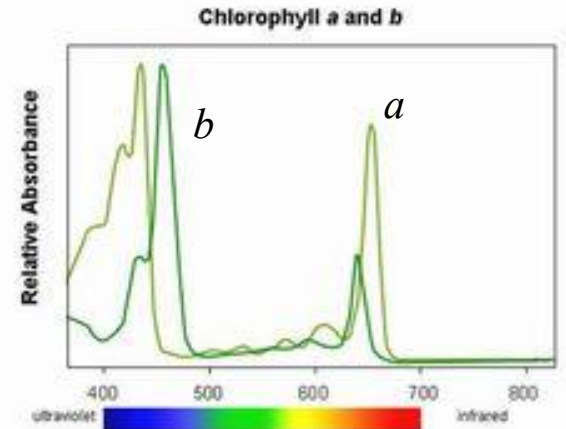
chlorophyll *b*



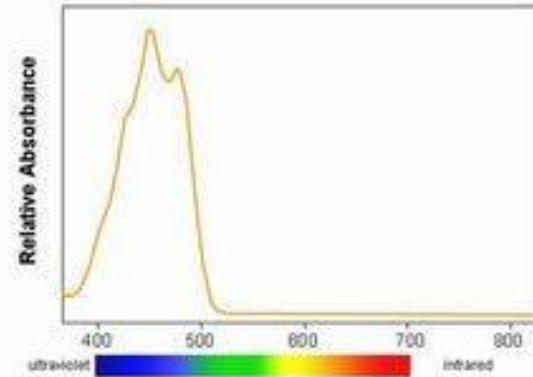
β-carotene



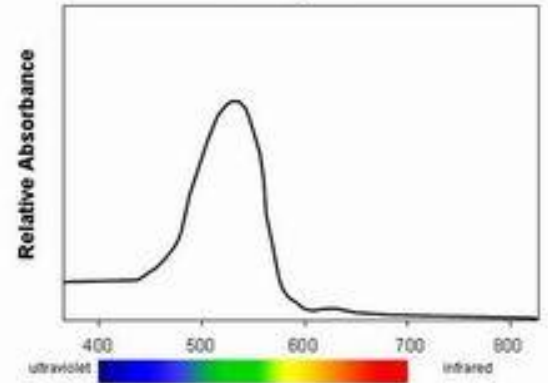
anthocyanane + glucose



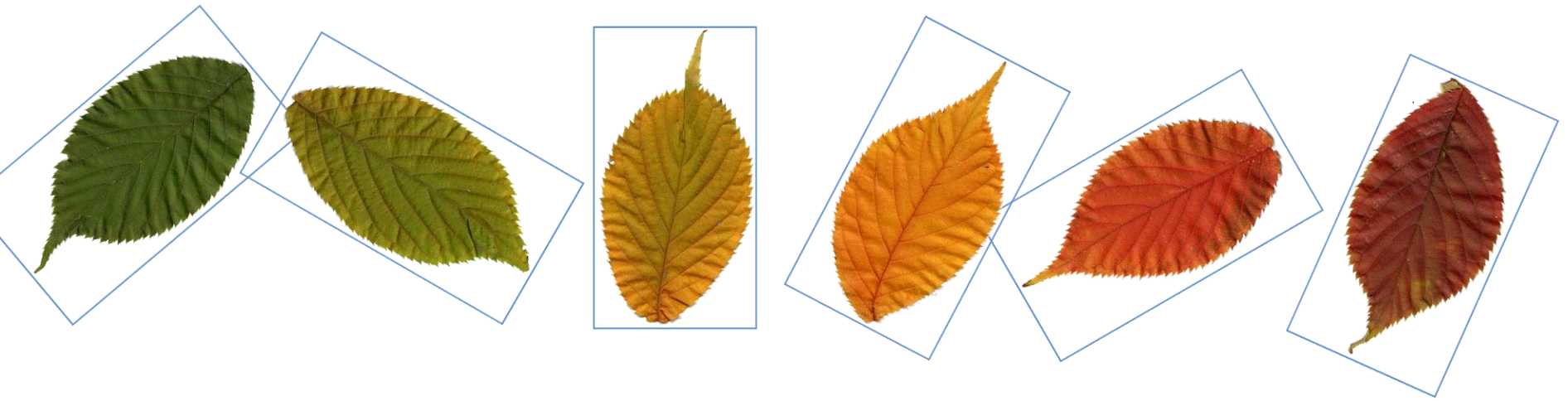
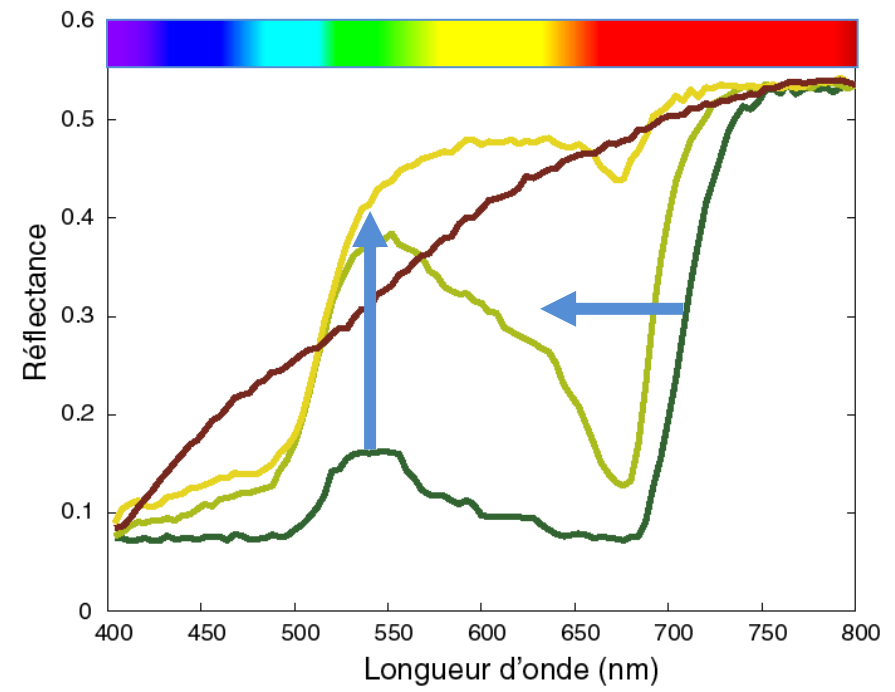
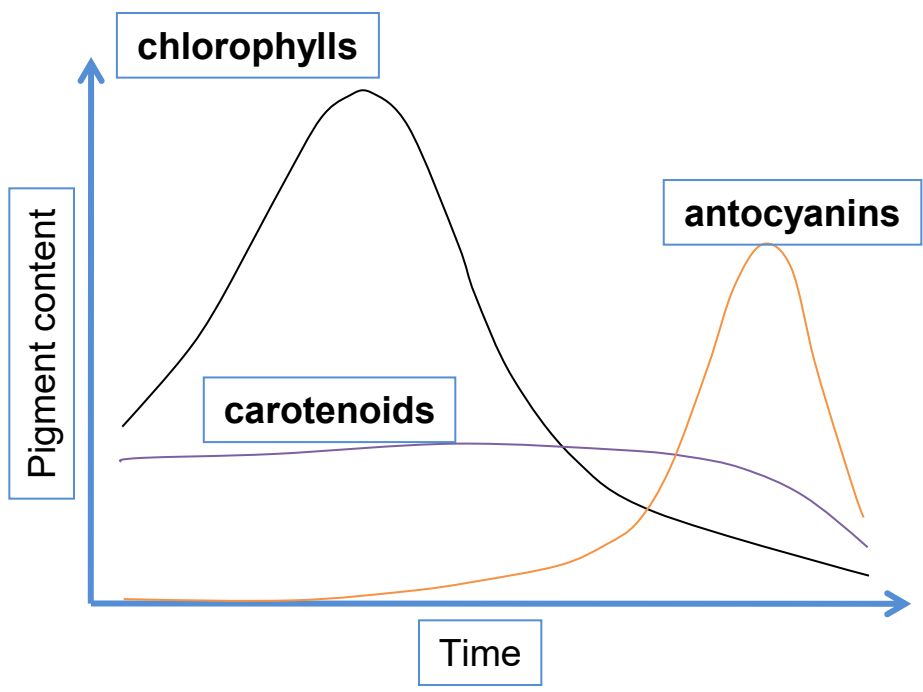
β -carotene



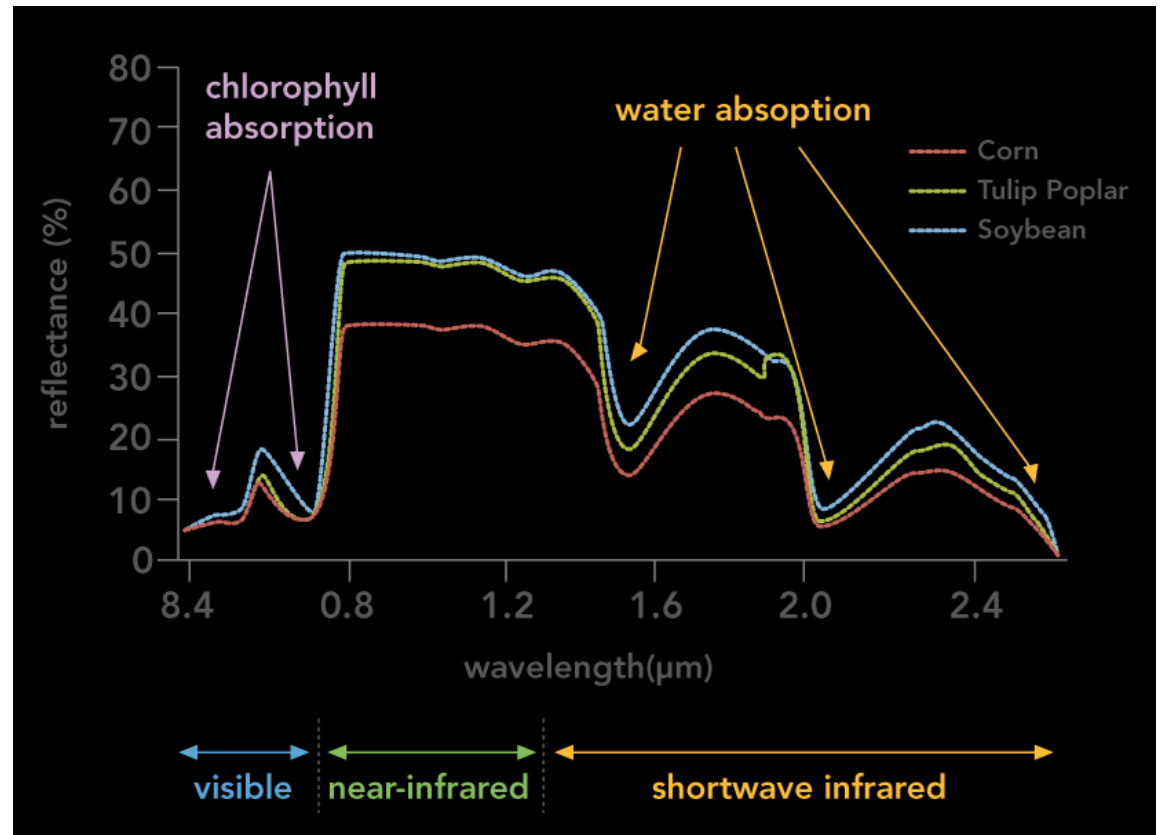
Anthocyanin



Vegetation: effect of leaf chemical composition



Vegetation: reflectance spectra



Vegetation: effect of the LAI

Temperate evergreen needleleaf forest

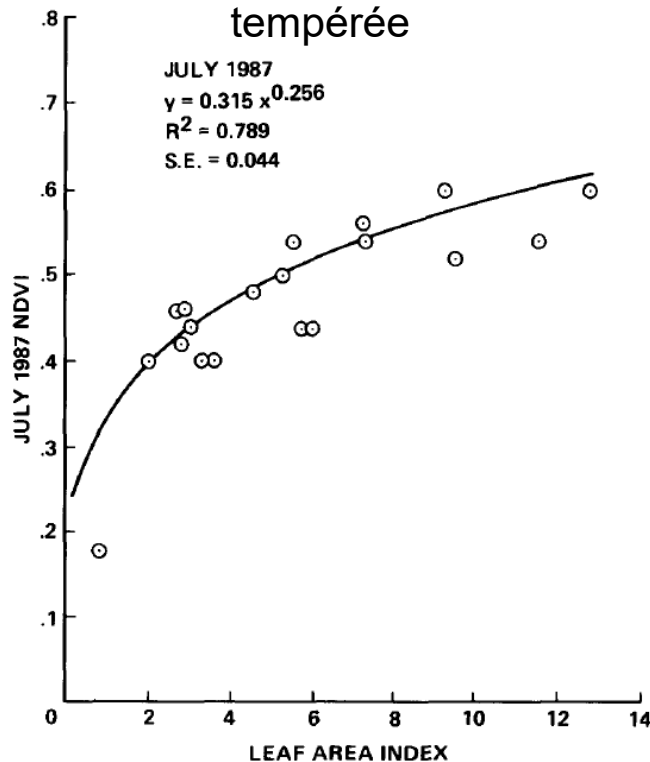


Figure 3. Relationship between leaf area index of the primary sites and the monthly maximum value composite NDVI for a) July 1986 and b) July 1987.

Soybean crop

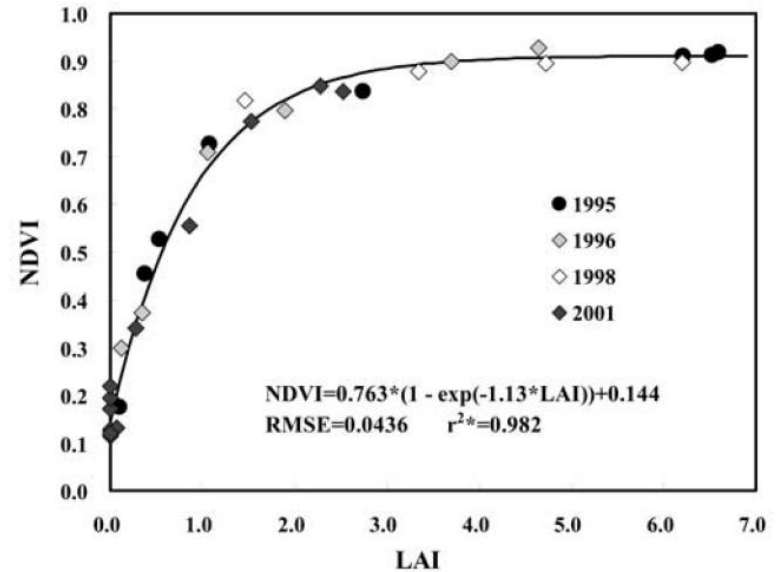
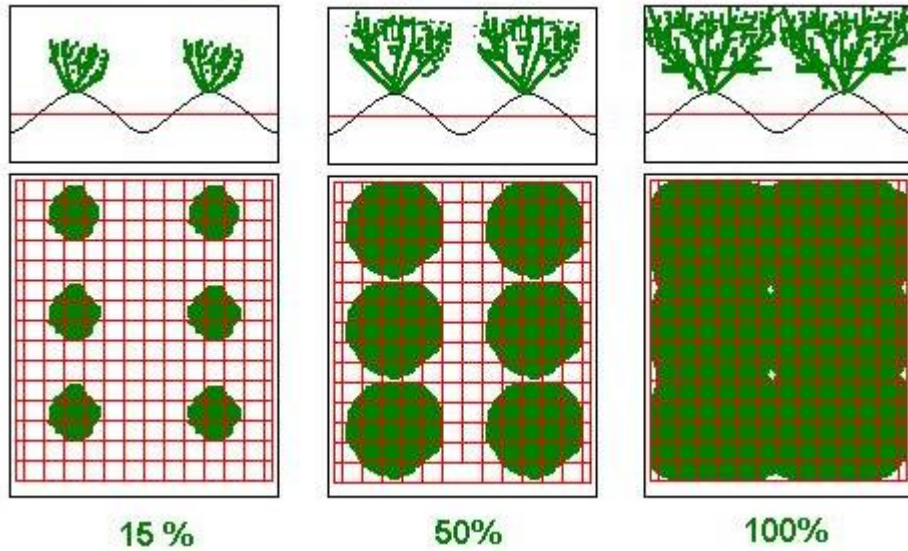


Figure 5. Relationship between leaf area index LAI and spectral vegetation index NDVI in soybean canopies derived from measurements in the 4 years: 1995, 1996, 1998 and 2001.

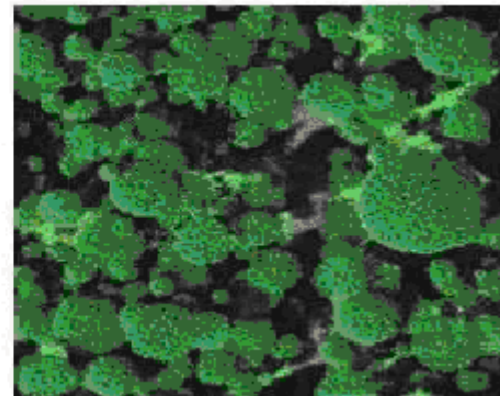
Inoue & Oliosio (2006), *Journal of Geophysical Research*, 111, D24S91.

Vegetation: fraction of ground cover (fCover)

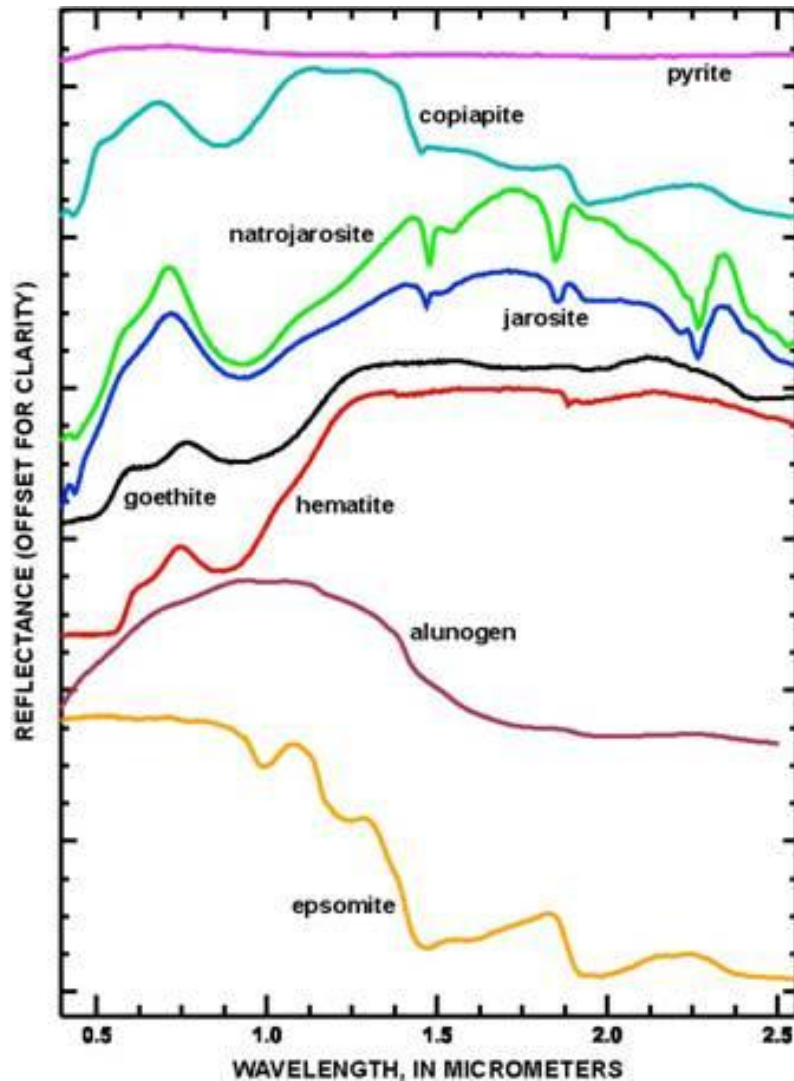
Crops



Forests



Bare soils: mineral optical properties



Hematite: $\text{Fe}_2\text{O}_3 \rightarrow$ oxide

Goethite: $\text{Fe}^{3+} \text{O} (\text{OH}) \rightarrow$ hydroxide

Jarosite: $\text{K Fe}^{3+}_3 (\text{SO}_4)_2 (\text{OH})_6 \rightarrow$ sulfate

Pyrite: $\text{Fe S}_2 \rightarrow$ sulfide



C.I. Grove, S.J. Hook & E.D. Paylor, 1992, *Laboratory reflectance spectra of 160 minerals, 0.4 to 2.5 micrometers*, JPL Publication 92-2.

<http://speclab.cr.usgs.gov/>

<http://minerals.gps.caltech.edu/>

Bare soils: mineral classification



Danxia Hills (China)

Bare soils: minerals + organic matter + water

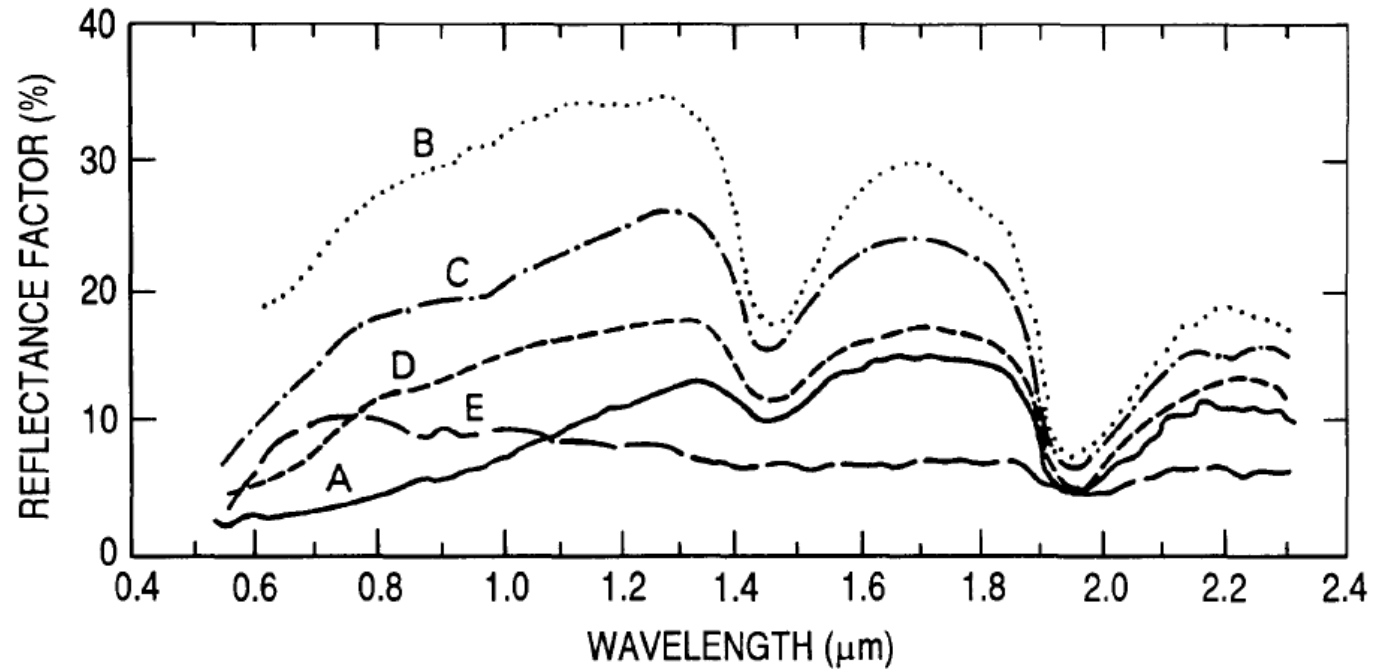
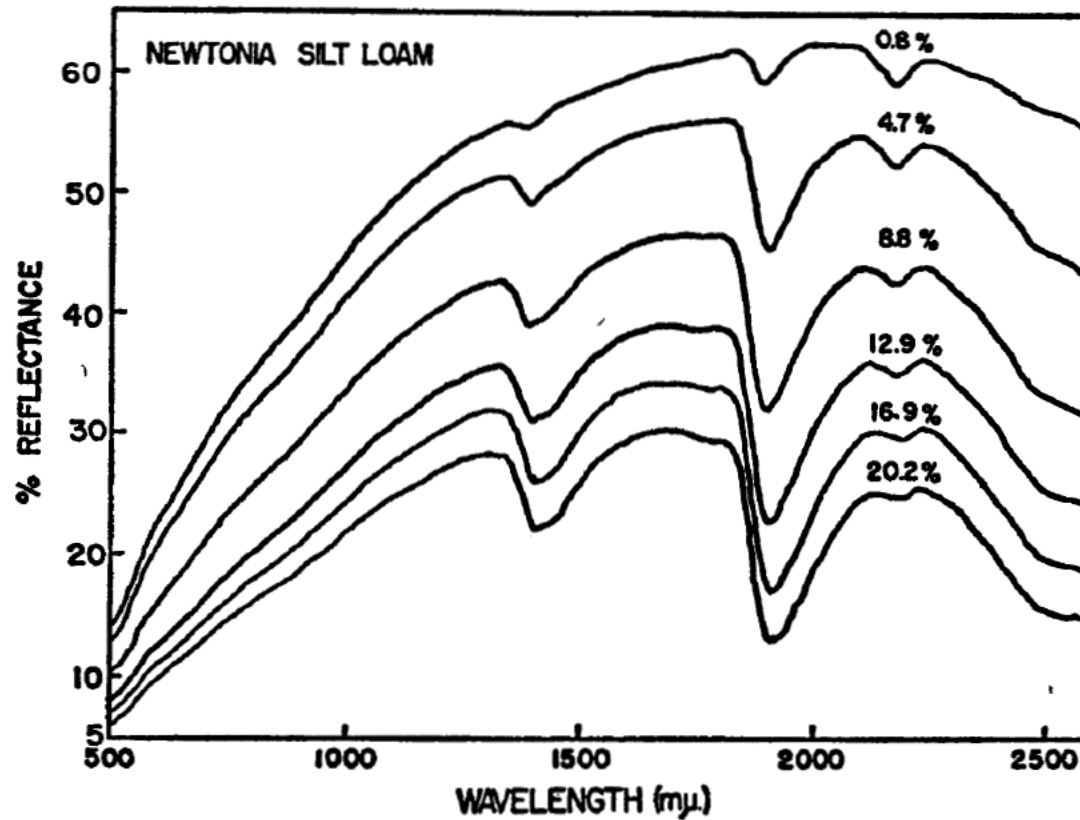
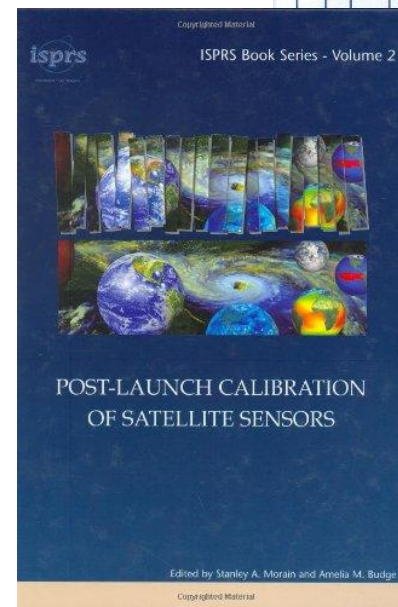
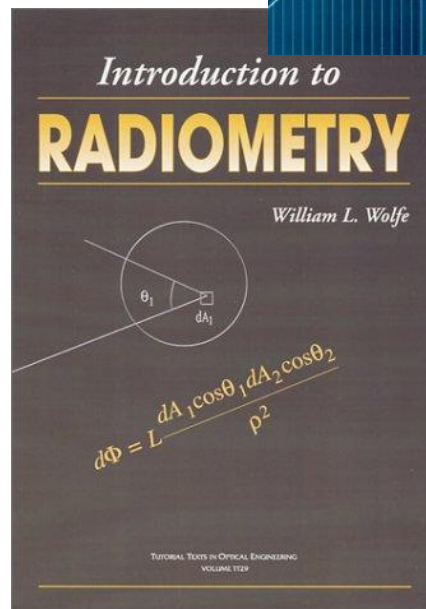
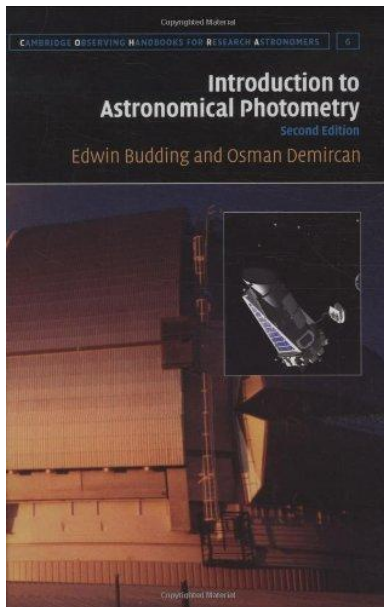
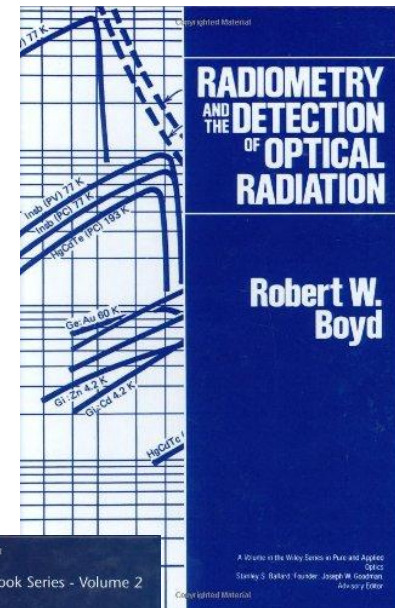
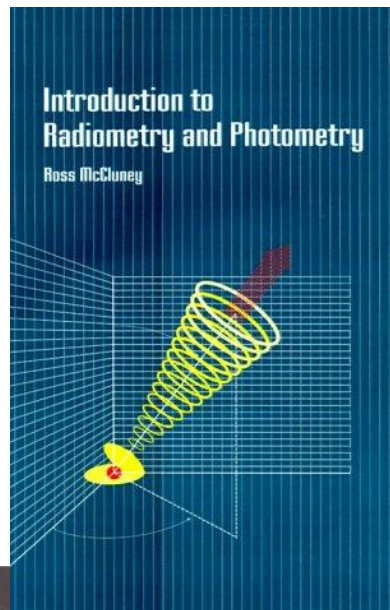
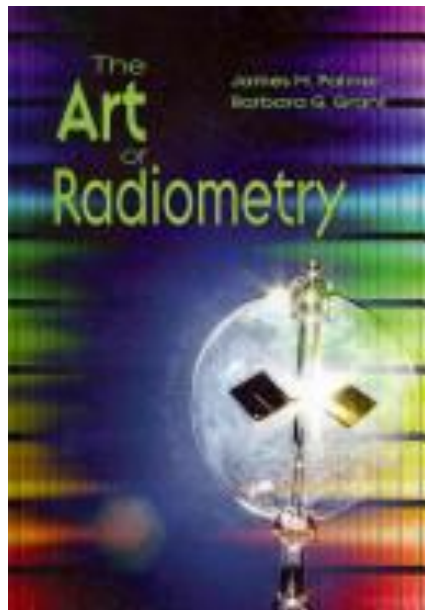


Figure 1. Figure taken from Stoner and Baumgardner (1981). Representative reflectance spectra of surface samples of five mineral soils: A) organic-dominated (high organic content, moderately fine texture); B) minimally altered (low organic, medium iron content); C) iron-affected (low organic, medium iron content); D) organic-affected (high organic content, moderately coarse texture); and E) iron-dominated (high iron content, fine texture).

Bare soils: effect of leaf water content



Platform	Sensor	Mode	Altitude	IFOV	Pixel size	Swath width
ER2	AVIRIS	HYPER	20 km	1 mrad	20 m	10,6 km
NOAA 5	AVHRR	MS	1 515 km	1,3 mrad	1,1 km	2 700 km
LANDSAT-7	ETM+	MS	705 km	0,04 mrad	30 m	183 km
		PAN		0,02 mrad	15 m	183 km
SPOT 6 & 7		MS	694 km	0,0086 mrad	6 m	60 km
		PAN		0,0022 mrad	1,5 m	
PARASOL	POLDER	MS	750 km		6 × 7 km	2400 km
Pléiades 1A		MS	694 km	0,003 mrad	2 m	20 km
		PAN		0,001 mrad	0.7 m	
TERRA	ASTER	VNIR	705 km		15 m	60 km
		SWIR			30 m	
		TIR			90 m	



III. Practical works

Instrument

Setting up system

Measurement steps

- Measuring the temperature of the light sources
- Measuring background ($\mu\text{W}/\text{cm}^2/\text{nm}$; *Micro-Watts per square centimeter per nanometer*)
- Measuring Reference
- Measuring reflectance (%)

Instrument

ASD FieldSpec 4



Wavelength range 350 nm – 2500 nm

Resolution VNIR 350nm - 700 nm 3 nm

Resolution SWIR: 1400 & 2100 nm 8 nm

Ocean Optic USB2000

Miniature Fiber Optic Spectrometer



Modular design

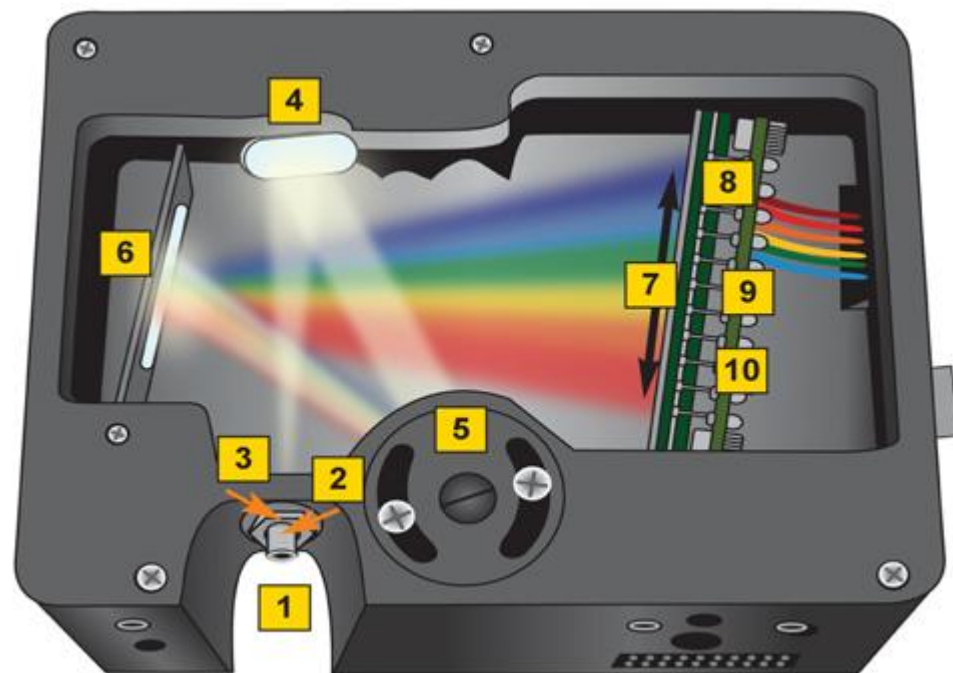
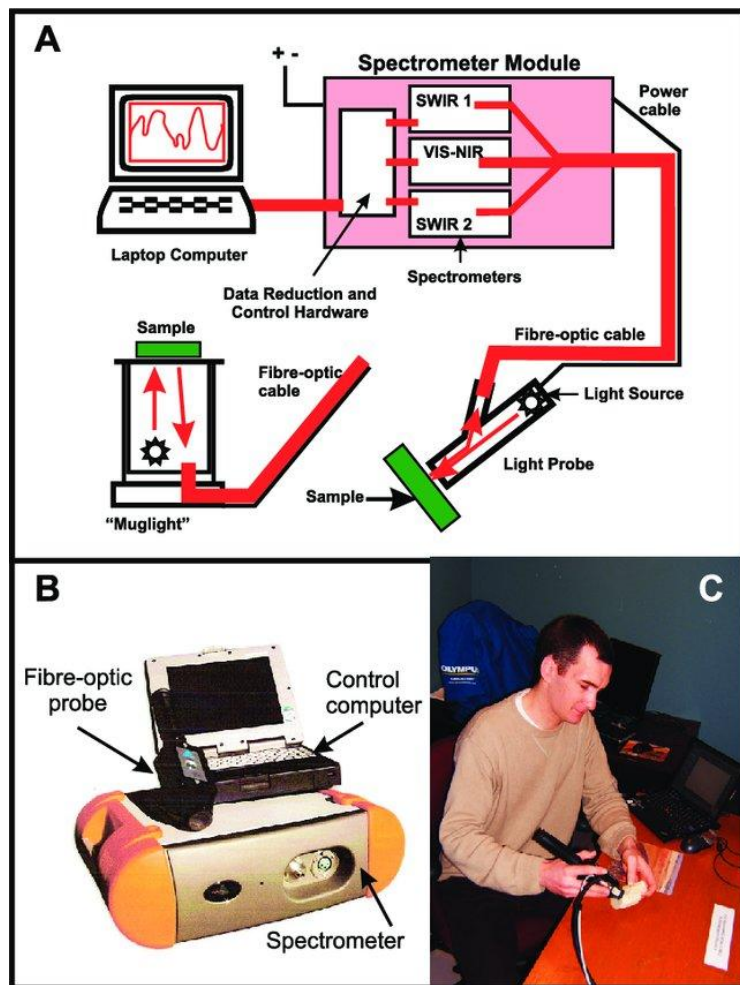
User-configured wavelength range
and resolution

Automatically reads the
wavelength calibration coefficients
of the spectrometer and configures
operating software

USB-to-PC interface; no external
power requirements

<http://www.oceanoptics.com/>

ASD FieldSpec 4 and Ocean Optics USB 2000: principle of measurement

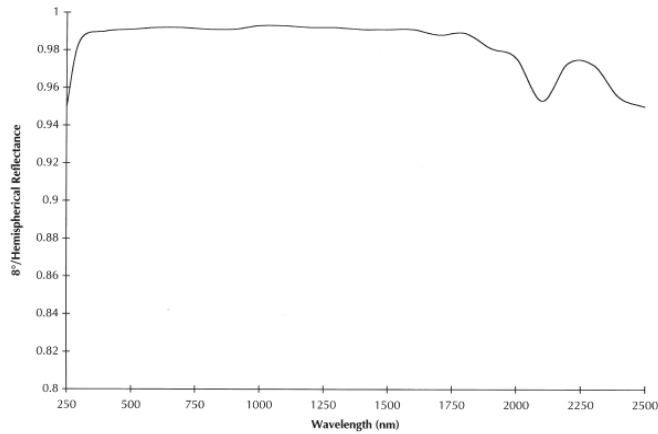
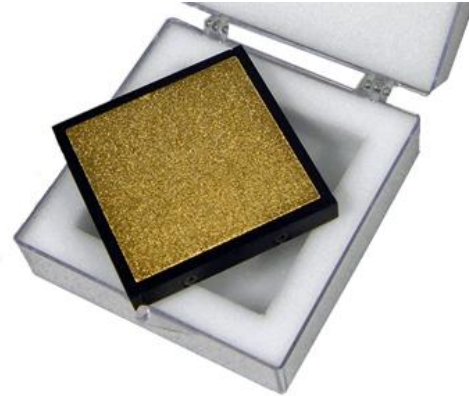


- 1-SMA 905 Connector
- 2-Fixed Entrance Slit
- 3-Longpass Absorbing Filter
- 4-Collimating Mirror
- 5-Grating & Wavelength Range
- 6-Focusing Mirror
- 7-L2 Detector Collection Lenses
- 8-Detector
- 9-OFLV Variable Longpass Order-sorting Filter
- 10-UV2 Detector Upgrades

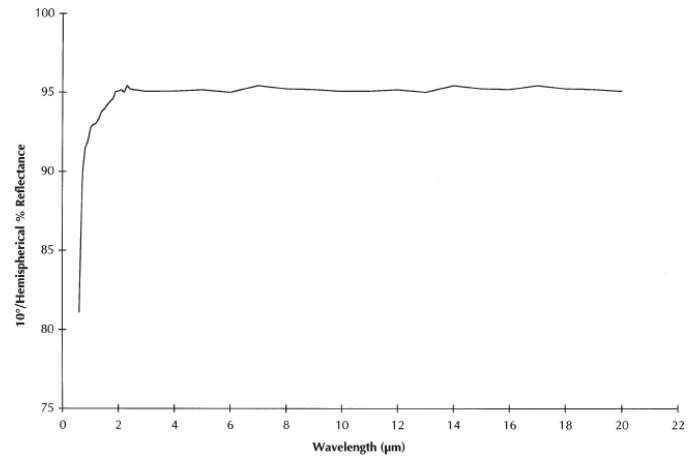
https://www.researchgate.net/publication/288899855_Visibleinfrared_spectroscopy_VIRS_as_a_research_tool_in_economic_geology_background_and_pilot_studies_from_New_Foundland_and_Labrador

http://www.oceanoptics.com/Products/benchoptions_usb4.asp

Perfect reflector: spectral optical properties (White body)

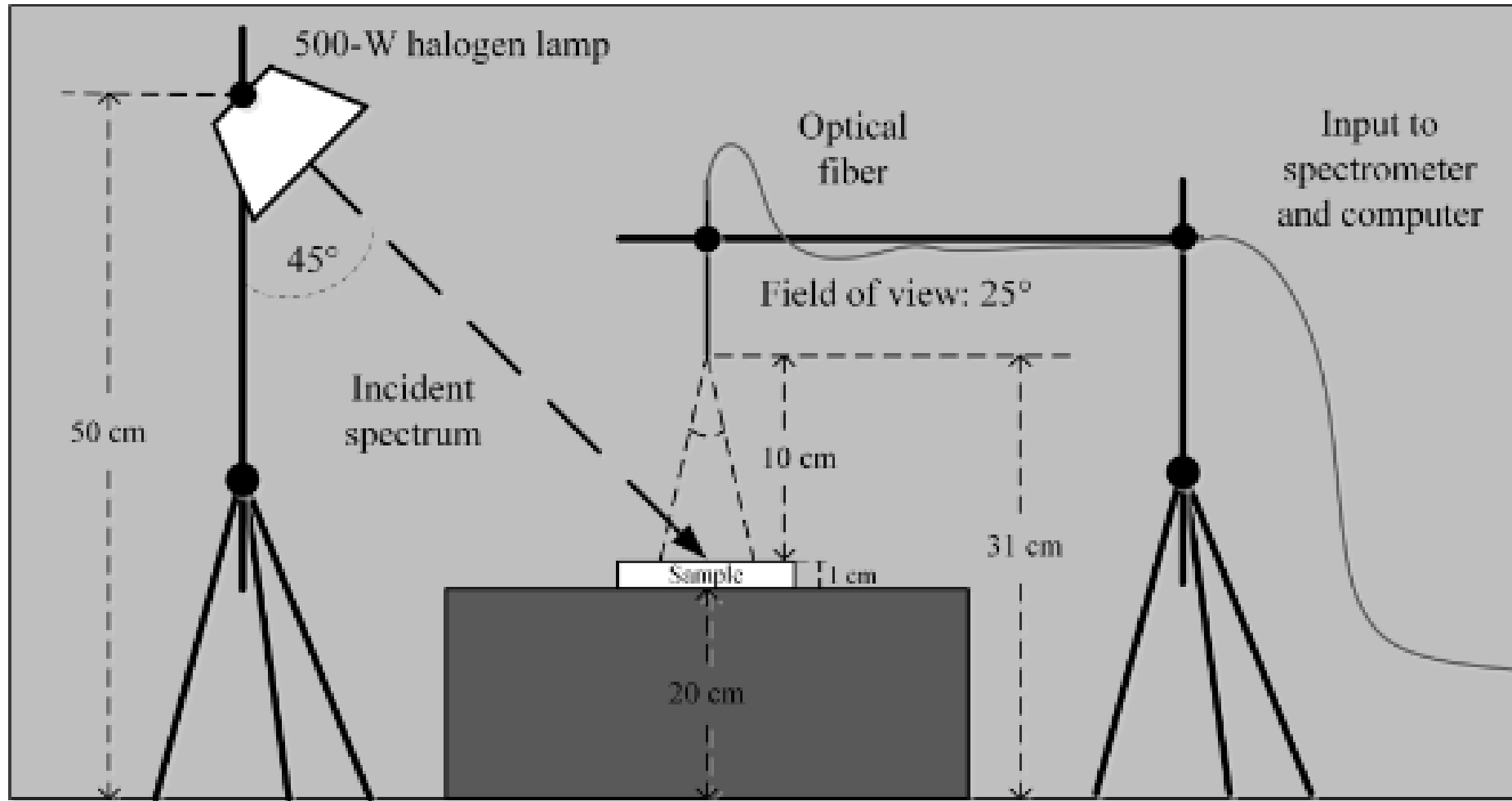


VIS-PIR-MIR : Spectralon



MIR-IRT : Infragold

Setting up the system



$$\%T_{\lambda} = \frac{S_{\lambda} - D_{\lambda}}{R_{\lambda} - D_{\lambda}} \times 100\%$$

Where:

S_{λ} = Sample intensity at wavelength λ

D_{λ} = Background intensity at wavelength λ

R_{λ} = Reference intensity at wavelength λ

Midterm: Presentations and Reports

- **Purpose:** Investigate the spectral reflectance of objects in the Lab and on the satellite images
 - Comparing the reflectance of different samples (dry leaves, green leaves, dry soil, water etc.)
 - Comparing the reflectance of samples with corresponded objects in satellite images.
 - Classification of satellite image to have landcover map in your study area (Using the field data related to the spectral measurement).
 - Writing report: max 10 pages (Upload reports and presentations to the Midterm report folder on the Moodle)
- **Structures of the Report:**
 - 1/ Introduction: The objectives? The requirement of your work? Why do you do your work? Tasks?
 - 2/ Data and method:
 - What types of data, material do you use? Description of your data, material.
 - Method: What tools, equipment, methods, working flows etc. do you use for the experiment? Characteristics of your method, tools, equipment?
 - 3/ Results:
 - Descriptions of your results: Figures, tables, statistical data.
 - 4/ Discussions:
 - Discussing about your results if they are good or not, any errors? Factors affect the accuracy of your results? How to improve your results?
 - What are the differences or the relation of satellite images and spectral measurement.
 - 5/ Conclusions:
 - Your main results? What do you get after your experiments? Any suggestions?
- **Structures of the Presentation**
 - 1/ Time frame: 15' for the presentation, and 10' for questions and discussions in the class.
 - 2/ Contents: Introduction, Data and methods, Results, Discussions, Conclusions